IITG RACING

Indian Institute of Technology Guwahati

Design and Development of Steering System for Formula Student Electric Vehicle

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1 Introduction

The complete prototype of steering system and its assembly is done in SolidWorks. To get that enhancement of steering response we majorly concentrated on Anti-Ackermann principle which is a primary consideration of a vehicle, based on it the vehicle behaviour depends. We are designing the steering geometry for formula cars and its components with a specified turning radius and wheel turning angles, the system could be optimized for a low steering effort.

This design overcomes the major issue of most of the formula student cars during endurance events i.e. driver's fatigue. While the lowest steering effort possible would be ideal, a compromise would have to be determined since a low steering effort also gave a slow steering response and therefore a large steering wheel angle at full lock. To allow for a fast steering response with a reasonable steering wheel angle, the full lock position was set to 360 degrees from lock-to-lock position. This compromise allowed for the uprights to incorporate the necessary moment arm to achieve the maximum wheel angle based upon the selected steering rack.

2 Design Considerations

2.1 Anti-Ackermann Geometry

Although Ackerman steering is efficient at low speeds it is not effective at higher speeds. This system incorporates a lower outer wheel angle and higher inner wheel angle. With increase in lateral forces there is a progressive increase in deflection i.e., the vehicle moves in a direction different to which it is actually pointing. At higher speeds lateral forces and vertical loads are significantly higher on the outer wheel causing a greater slip angle, due to which the Ackerman system does not perform well as the inner wheel is subjected to higher slip angles causing increase in temperature and also produces a drag.

The additional reasons for choosing Anti-Ackermann are as follows:

- The rack and pinion arrangement is placed ahead of the front axle and likewise the steering arm points towards the front of the vehicle — the outer wheel makes a smaller radius than the inner wheel while taking a turn, which is opposite to that of Ackerman steering.
- To get a better steering response by reducing slip and providing more grip to the outer tire during cornering at larger radius turns.

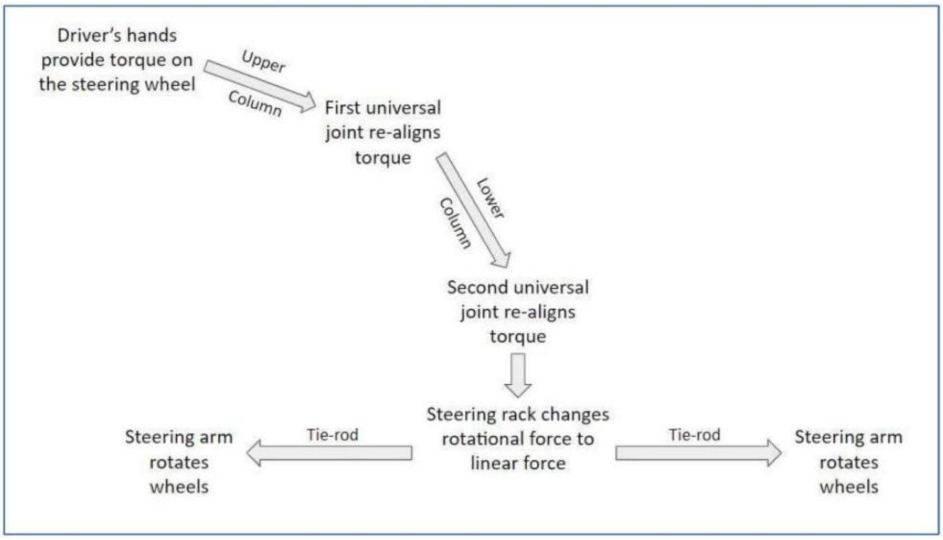


Figure 1: Steering Force Flowchart

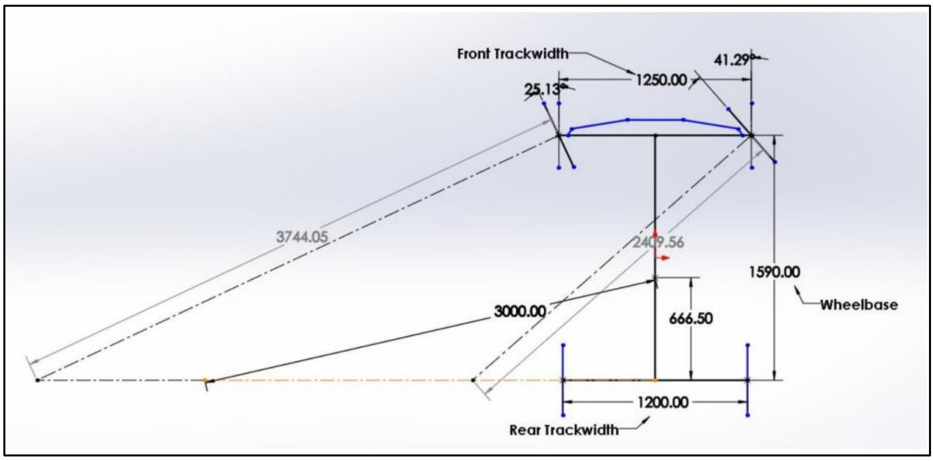


Figure 2: Anti-Ackermann Steering Geometry

2.2 Rack and Pinion Design

In rack and pinion steering mechanisms, the steering wheel turns the pinion gear; the pinion moves the rack, which is a linear gear that meshes with the pinion, converting circular motion into linear motion along the transverse axis of the car (side to side motion). This motion applies steering torque to the kingpin of the steered wheels via tie rod and a short lever arm called the steering arm. The rack and pinion design has the advantages of a large degree of feedback and direct steering "feel"; it also does not normally have any backlash or slack. That is why this system is a racing designer's first choice. A disadvantage is that it is not adjustable — when it does wear and develop lash, the only cure is replacement.

Some advantages of this system are as follows:

- Economical and simple to manufacture.
- Easy to operate due to good degree of efficiency.
- Contact between steering rack and pinion is free of play and even internal damping is maintained.

- The idler arm (including bearing) and the intermediate rod are no longer needed.
- Tie rods can be joined directly to the steering rack.
- Easy to limit steering rack travel and therefore the steering angle.

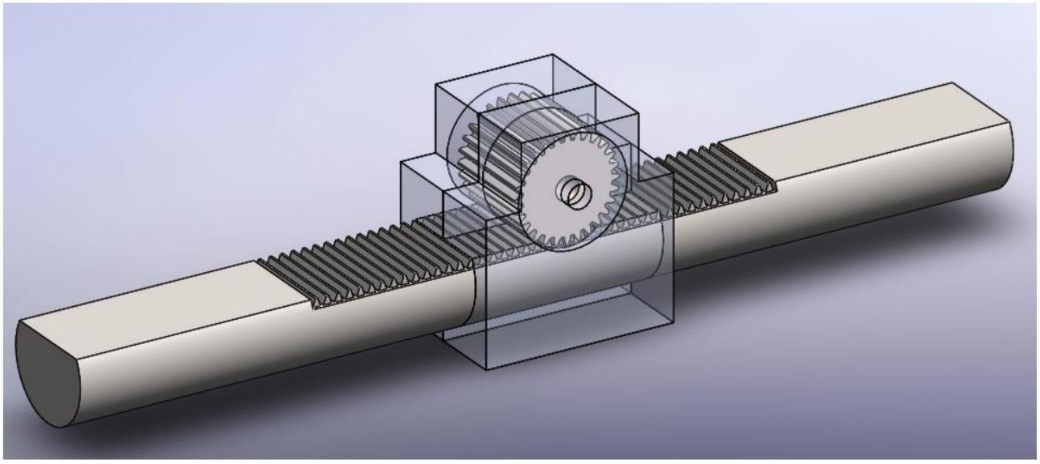


Figure 3: Rack and Pinion Assembly

2.3 Steering Wheel

An ergonomically designed steering wheel of 1.9 cm diameter is made of laser-cut aluminum to reduce the steering effort of the driver. As a requirement of the design for the competition, the steering wheel must be equipped with a quick-release mechanism. Following discussion with the team it was decided to make use of the last year quick release mechanism.

Figure 4: Steering Wheel Assembly

3 Calculations

3.1 Anti-Ackermann Calculations

The sketch for Anti-Ackermann Geometry was drawn for the car in SOLIDWORKS keeping in mind the rules of Formula Bharat 2026. Keeping the distance of CG from IC as 3000 mm (turning radius), the inner and outer wheel angles were obtained as:

$$\theta_i = 21.53° \qquad \theta_o = 22.8°$$

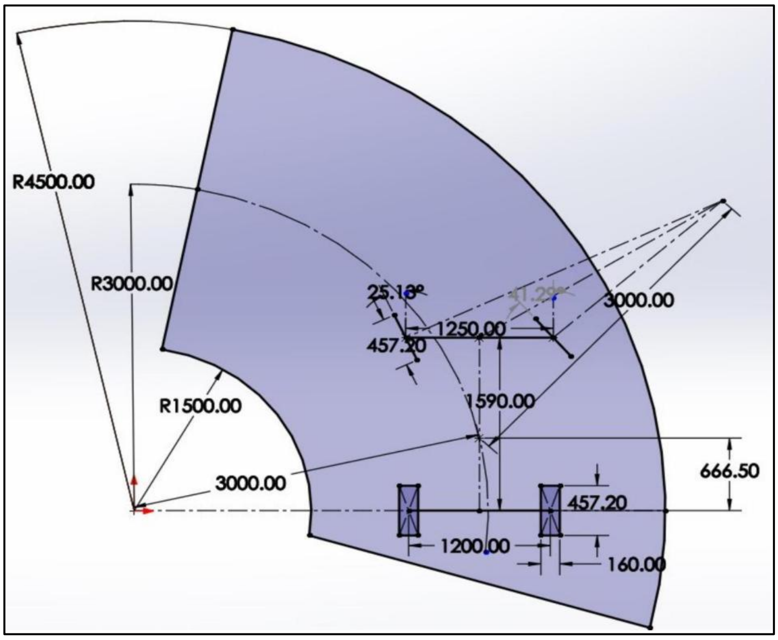


Figure 5: Turning Radius Geometry for Anti-Ackermann Condition

The design specifications are listed below:

Sl. No	Parameters	Values
1	Wheelbase	1590 mm
2	Front Trackwidth	1250 mm
3	Rear Trackwidth	1200 mm
4	Tire Width	160 mm
5	Tire Radius	9 inches
6	Track Turning Radius	3000 mm

Table 1: Design Specifications

3.2 Ackermann Angle

Let φ be the Ackermann angle. Using the formula:

$$\tan \varphi = (\text{kingpin pivot–to–pivot distance}) \,/\, (2 \times \text{wheelbase})$$
$$\tan \varphi = 1147.70 \,/\, (2 \times 1590) = 0.3609$$
$$\varphi = 14.00°$$

3.3 Space Required for Turning

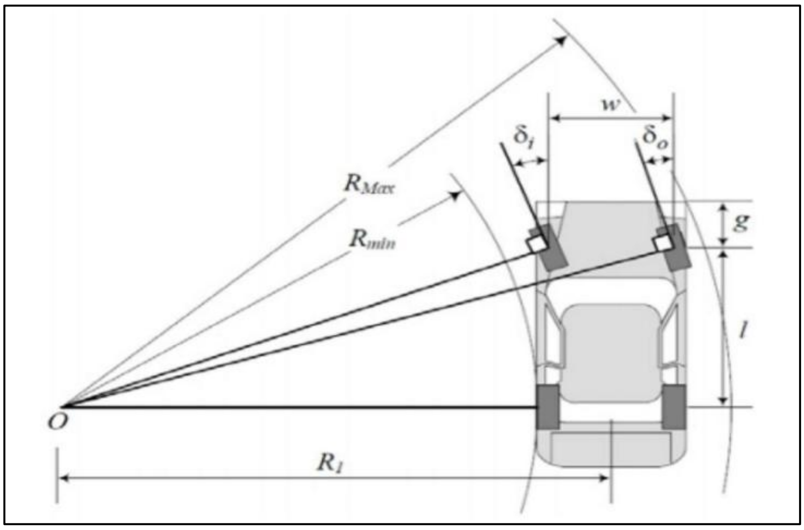


Figure 6: Turning Space Geometry

The space required for turning is the space between the two circles in which the whole vehicle fits without going out of the circle. It is given by:

$$\Delta R = R_{max} - R_{min}$$

R_{max} and R_{min} are obtained from the formulas below (g = distance from front axle to nose):

$$R_{min} = l / \tan\theta_i = 1590 / \tan(25.13°) = 3389.66 \text{ mm}$$

$$R_{max} = \sqrt{[(R_{min} + w)^2 + (l + g)^2]}$$

$$R_{max} = \sqrt{[(3389.66 + 1250)^2 + (1590 + 600)^2]} = 5130.55 \text{ mm}$$

$$\Delta R = R_{max} - R_{min} = 1740.89 \text{ mm} \approx 1.75 \text{ m}$$

3.4 Turning Radius

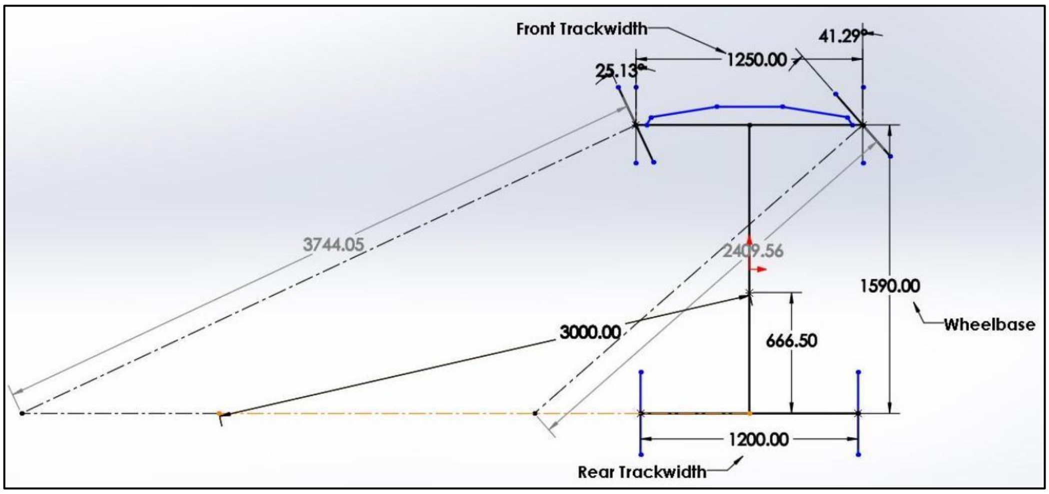


Figure 7: Top View of Steering Geometry

Using the wheel angles computed above, each turning radius is calculated as:

$$R_{if} \text{ (Inner Front)} = \text{Wheelbase} / \sin\theta_i = 1590 / \sin(21.53°) = 3744.05 \text{ mm}$$

$$R_{of} \text{ (Outer Front)} = \text{Wheelbase} / \sin\theta_o = 1590 / \sin(22.8°) = 2409.56 \text{ mm}$$

$$R_{ir} \text{ (Inner Rear)} = \text{Wheelbase} / \tan\theta_i = 1590 / \tan(21.53°) = 3389.67 \text{ mm}$$

$$R_{or} \text{ (Outer Rear)} = \text{Wheelbase} / \tan\theta_o = 1590 / \tan(22.8°) = 1810.50 \text{ mm}$$

3.5 Tie Rod Calculation

Defining the variables used in the tie rod calculation:

- x = steering arm length
- y = tie-rod length
- z = rack ball joint center to ball joint length
- d = distance between front axis and rack center axis
- B = distance between left and right kingpin centerline
- β = Ackermann angle

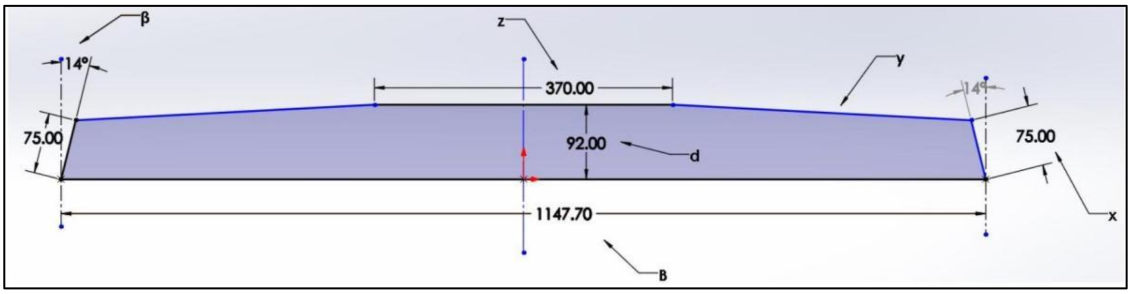


Figure 8: 2D Sketch of Steering Rack Assembly in Top View

From the figure above:

$$y = \sqrt{ [(B - z) / 2 - x\cdot\sin\beta]^2 + [d - x\cdot\cos\beta]^2 }$$

The tie-rod length came out to be approximately 391.72 mm.

3.6 Rack Travel

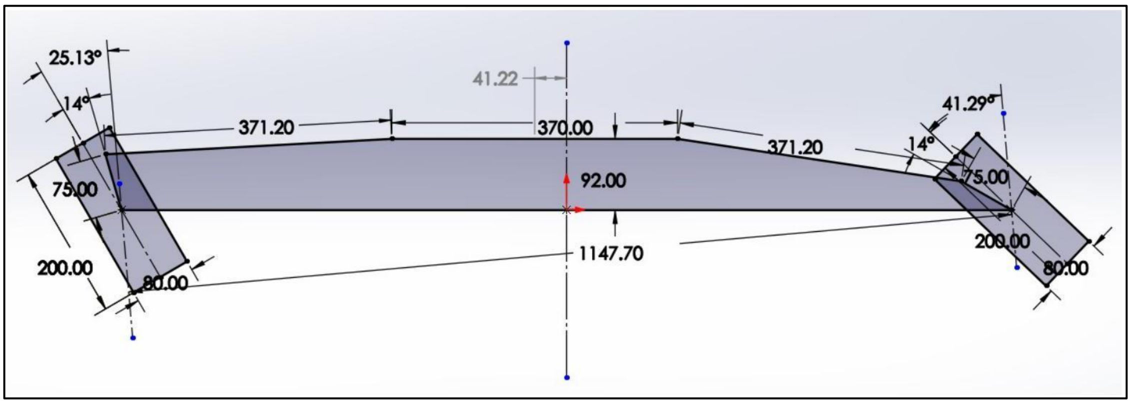


Figure 9: Rack Travel for Maximum Turn Angles

Rack travel δ for maximum wheel turn angle was found in SOLIDWORKS:

$$\delta = 30.30 \text{ mm}$$

3.7 Steering Ratio

The steering ratio is calculated as:

$$\text{Steering Ratio} = (\text{Angle of steering wheel}) / (\theta_i + \theta_o)$$
$$= 180 / (21.53 + 22.8) = \mathbf{4.06}$$

3.8 C-Factor

C-factor is the linear distance travelled by the rack in one rotation of the pinion:

$$\text{C-factor} = \text{Total Rack Travel (mm)} / \text{Lock-to-lock angle (rad)}$$

$$\text{Lock-to-lock angle} = \theta_i + \theta_o = 21.53 + 22.8 = 44.33° = 0.77 \text{ rad}$$

$$\text{C-factor} = 2 \times 41.22 / 1.16 = 107.66 \text{ mm}$$

3.9 Pinion Calculation

It is assumed to get maximum rack travel at 180° of steering wheel rotation. The maximum rack travel being 30.30 mm, the pitch diameter of the pinion is found using the arc length formula. Assuming 90% efficiency to account for play in universal joints and steering column:

$$s = r \times \omega$$
$$30.30 = r \times 0.9 \times \pi$$
$$r = 30.30 / (0.9 \times \pi) = 10.71 \text{ mm}$$
$$PCD = 2r = 21.42 \approx 22 \text{ mm}$$

Therefore: Module M = 2, PCD = 22 mm, Number of teeth on pinion T_p = PCD / M = **11**, Circular Pitch = $\pi \times M$ = 6.28 mm.

3.10 Rack Calculation

Total rack travel for both sides = 2δ = 60.60 mm. Module M = 2 (same as pinion for perfect meshing). Axial pitch = $\pi \times M$ = 6.28 mm. Number of teeth on rack = $2\delta / (\pi M) \approx$ **11** (with clearance). Minimum actual rack length:

$$L_r = T_r \times \pi \times M = \mathbf{62.83 \text{ mm}}$$

3.11 Steering Effort Calculation

Given: Mass of car M = 310 kg, weight distribution 45% front / 55% rear. Force on any front wheel:

$$W = ½ \times (45/100) \times 310 \times 9.81 = 684.25 \text{ N}$$

With coefficient of friction μ = 1.4 and lateral coefficient μ_l = 2.1:

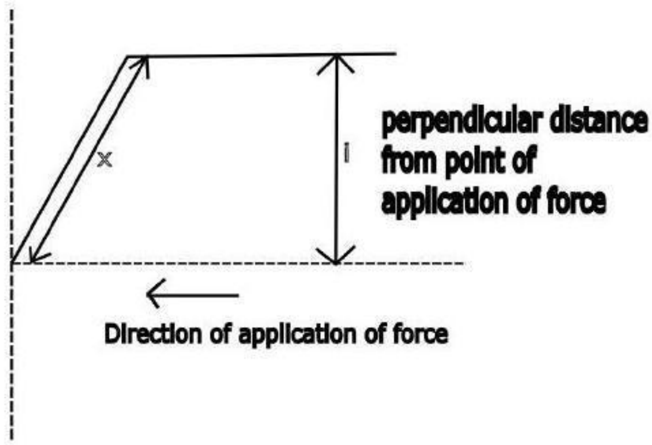
$$\text{Frictional Force } F_r = \mu \times W = 957.95 \text{ N}$$
$$\text{Lateral Force } F_y = \mu_l \times W = 1436.92 \text{ N}$$

Torques on the wheel (Scrub Radius S_r = 30 mm, Mechanical Trail M_r = 9.22 mm):

$$T_R \text{ (Resisting)} = F_r \times S_r = 957.95 \times 30 = 28{,}738.50 \text{ N·mm}$$
$$T_L \text{ (Lateral)} = F_y \times M_r = 1436.92 \times 9.22 = 13{,}254.50 \text{ N·mm}$$
$$T \text{ (Total)} = T_L + T_R = 41{,}993 \text{ N·mm}$$

Force on steering arm by tie-rod P (perpendicular distance i = 92 mm):

$$P = T / i = 41{,}993 / 92 = 456.44 \text{ N}$$



From suspension geometry, tie rod inclination $\alpha = 9.64°$. Therefore:

$$\text{Force on Rack} = P \times \cos(\alpha) = 456.44 \times \cos(9.64°) = 450 \text{ N}$$
$$\text{Torque on Pinion} = \text{Force on rack} \times \text{pinion radius} = 450 \times 15 = 6{,}750 \text{ N·mm}$$

Assuming 90% efficiency (η) for the universal joints, minimum required torque at steering wheel T_{Smin}:

$$T_{Smin} = \text{Torque on pinion} / \eta = 6{,}750 / 0.9 = 7{,}500 \text{ N·mm}$$
$$\text{Steering Effort} = \text{Torque on steering wheel} / \text{Steering wheel radius} = 7.5 / 0.5 = 15 \text{ N}$$

3.12 CAD Model of Steering Assembly

The complete steering assembly was modelled in SOLIDWORKS. The rendered version is shown below.

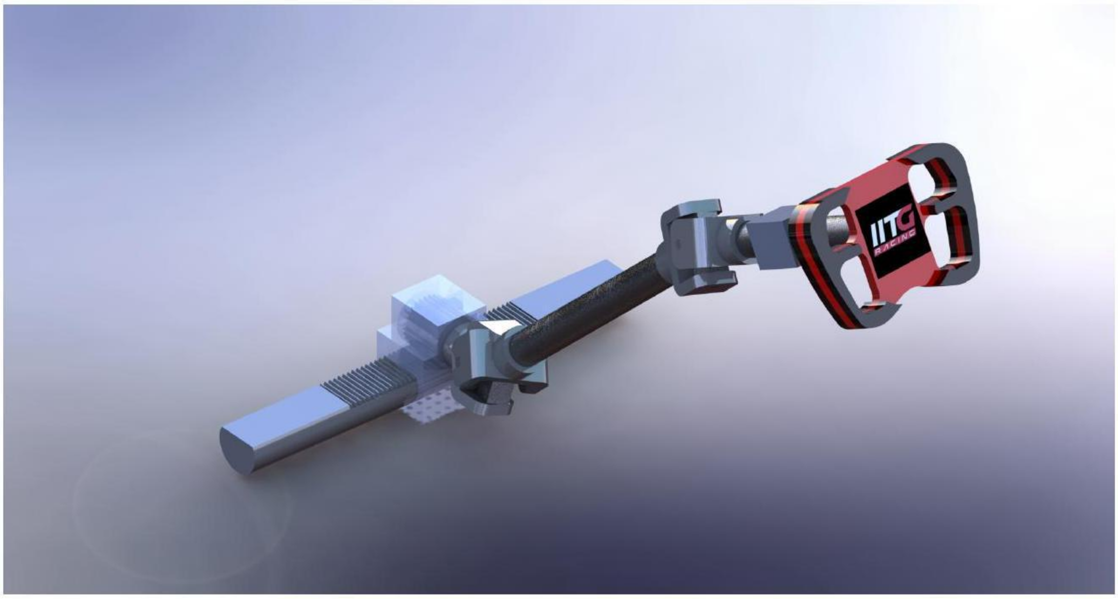


Figure 10: Steering Wheel Assembly